D031: Editorial apparatus

This apparatus accompanies the original French and standalone English editions. Source authority is the supplied IAS 43-page witness. Its article occupies printed pages 247-289. The publication header cites 247-290, but neither supplied witness contains page 290; none is invented.

The normalized readers retain a page boundary for every physical source leaf and preserve all native mathematical code. Page numbers refer to the original printed pages. Running heads, folios and scanner/copy matter are recorded below, not inserted into the article body. The inherited salvage archive and its branches remain ZERO_ACCEPTED; they are not a source of accepted text.

Typesetting normalization

The new TeX and PDF readers are editable mathematical editions, built from the exact supplied French and English page records. Markdown page headings and operational status prose are excluded. The original contents lists are rendered as native tables, formulas remain math, and all 23 commutative diagrams in each language remain native TikZ-CD. Reader page-to-source identity is checked after compilation. No image fallback is used; therefore no crop derivatives or unlogged image substitutions are introduced.

Provenance-only comparisons against inherited candidate material are preserved in the unmodified input apparatus and salvage ledger, but are not promoted into this reader-facing apparatus. Omitted inherited note identifiers: A003, A006, A014, A017, A019, A025, A028, A031, A034, A037, A040, A044, A048, A051, A054, A057, A060, A063, A066, A069, A072, A075, A078, A081, A084, A087, A090, A093, A096, A099, A102, A105, A108, A111, A114, A117, A120, A123, A126, A130. The source authority remains controlling.

Printed page 247 (physical 1)

A001 - PAGINATION_DISCREPANCY

The proceedings header reads “Vol. 33 (1979), part 2, pp. 247-290,” while both exact witnesses contain 43 physical leaves ending at printed 289.

Preserve the 247-290 citation claim in apparatus; map and edit only printed 247-289; never create page 290.

A002 - COPY_MATTER_EXCLUSION

The authority includes the proceedings citation line, AMS(MOS) subject classifications, copyright line, folio, scanner border, and a neighboring-page sliver.

Exclude these from the reader body; retain the exact title, byline, Sommaire, and article text; record the excluded copy matter here.

Printed page 248 (physical 2)

A004 - COPY_MATTER_EXCLUSION

The authority leaf carries a repeated running head and folio, a black scanner border, and a substantial neighboring-page sliver at right.

Exclude all such copy matter and sliver text; preserve only the owned-page article text.

A005 - TRANSLATION_DECISION

The source terms are “domaine hermitien symétrique,” “dual D-check,” and “domaine de Siegel de 3e espèce.”

Translate as “Hermitian symmetric domain,” retain the symbol D-check, and use “Siegel domain of the third kind.”

Printed page 249 (physical 3)

A007 - DIAGRAM_GATE_F01

The authority diagram has objects C(F), C(E), and G(A)/rho G-tilde(A) · G(Q), with both diagonal arrows and a vertical C(F) to C(E) arrow labeled N_{F/E}; the candidate omits the vertical norm arrow.

Recreate the full diagram in native tikz-cd from authority pixels; freeze the vertical N_{F/E} arrow and both diagonals.

A008 - TRANSLATION_DECISION

The technical terms are “corps dual,” “modèle faiblement canonique,” and “groupe des classes d’idèles.”

Translate respectively as “reflex field,” “weakly canonical model,” and “idèle class group”; preserve all symbols independently against the authority.

A009 - NOTATION_CHECK

The authority uses rho:G-tilde to G, G(A^f), M^0_{Q-bar}(G,X+), $G^{\mathrm{ad(R)}}$+, $G^{\mathrm{ad(Q)}}$+, and Z(Q)^-.

Preserve these notations and the displayed pi_0 quotient equality; reject OCR substitutions A-prime and unbarred Q.

D031-N001 - SOURCE-FAITHFUL DIAGRAM GEOMETRY

Independent inspection of authority physical page 3 / printed page 249 found that both arrows to the quotient are oblique: the quotient is vertically centered between C(F) and C(E). The returned Markdown placed the quotient on the lower row and made the lower arrow horizontal. Both staged readers restore the centered quotient and rising lower arrow using a three-row native TikZ-CD diagram. The vertical norm arrow, endpoints, labels, and algebraic expressions are unchanged. The input-state bytes remain untouched.

Printed page 250 (physical 4)

A010 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, black border, and neighboring-page sliver; the owned text begins by completing “l’ac-tion.”

Exclude copy matter and the sliver; preserve the page-boundary continuation and all section 0.1 text through “Nous n’utiliserons.”

A011 - TRANSLATION_DECISION

The source calls the central-modification method “maladroite,” proposition [5, 1.15] “sybilline,” and speaks of a “provision de modèles canoniques.”

Translate as “awkward,” “cryptic,” and “a supply of canonical models,” while keeping the restrained first-person tone.

Printed page 251 (physical 5)

A012 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, scanner border, and neighboring-page sliver; article text continues from physical page 4.

Exclude copy matter and sliver text; preserve the continuation and the exact openings of 0.2-0.8 and Section 1.

A013 - NOTATION_CHECK

The authority distinguishes superscript 0, superscript +, and subscript +; defines A^f as a restricted product; displays pi_0 A_$E^{*/E*}$ to Gal(Q-bar/E)^ab; and numbers h(z)v=$z^{\text{\{-p\}z-bar}}${-q}v as (1.1.1.1).

Preserve all distinctions, the displayed reciprocity map, and the equation number in French and English.

Printed page 252 (physical 6)

A015 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, black scanner border, and neighboring-page sliver; article text begins by completing “ex-tension.”

Exclude copy matter and sliver text; preserve the continuation and stop exactly after item 1.1.7(a).

A016 - NOTATION_CHECK

The authority defines w_h as the restriction of h^{-1}, defines mu_h by z^{-p}, uses $F^{\text{p=sum\_\{r>=p\}V}}${rs}, type script-C, and Z(n)=dfn Z(1) tensor n.

Preserve each sign, exponent, index, and definition marker in both layers.

Printed page 253 (physical 7)

A018 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, a black scanner border, and a neighboring-page sliver; the owned text begins with item 1.1.7(b) and ends mid-sentence at h:S→.

Exclude the running head, folio, border, and sliver; preserve the exact page-boundary continuation into physical page 8.

A020 - TRANSLATION_DECISION

The source terms are “transversalité,” “variation de Z-structure de Hodge,” and “polarisation.”

Translate as “transversality,” “variation of Z-Hodge structure,” and “polarization,” preserving the displayed operators and Tate twists.

Printed page 254 (physical 8)

A021 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, a black scanner border, and a neighboring-page sliver; it continues the sentence from physical page 7 and ends immediately before the displayed tangent-space diagram.

Exclude copy matter and sliver text; preserve the continuation and stop exactly after "dφ est le composé."

A022 - NOTATION_CHECK

The authority uses h(R*) rather than w(R)* in 1.1.13(α), gives V_{iC}$^{n=\Sigma\_\{p+q=n\}V\_i}${pq}, names the Grassmannian map φ, and names the representation map ρ:L→End(V).

Preserve h(R*), the full graded-piece formula, φ, and ρ in French and English; reject candidate substitutions and its simplified proof diagram.

Printed page 255 (physical 9)

A023 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, a black scanner border, and a neighboring-page sliver; article text begins with the tangent-space diagram and ends after the opening of reduction step (2).

Exclude copy matter and sliver text; preserve the complete diagram and the page-boundary continuation to physical page 10.

A024 - DIAGRAM_AUDIT

The authority diagram is a commutative square with L/L^{00}, End(V)/End(V)^{00}, L_C/F^0L_C, and End(V_C)/F^0End(V_C); horizontal arrows are labelled ρ and vertical arrows i.

Recreate the full square in native tikz-cd with every object and label; check the English diagram independently against the authority.

Printed page 256 (physical 10)

A026 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, a black scanner border, and a neighboring-page sliver; the owned text begins by completing "sous-groupe central" and ends with the final sentence of 1.1.18(b).

Exclude copy matter and sliver text; preserve both page-boundary continuations.

A027 - SOURCE_ANOMALY

The authority literally prints "un produits d'espaces," h(z)=m(z/z-bar) after introducing m_x(u), h(U_1) although physical page 9 has U^1, and the final target Q(n) after first discussing forms valued in Q(-n).

Keep every literal reading in the frozen French. In English, regularize only ordinary grammar, preserve the displayed symbols h(z)=m(z/z-bar), U_1, and Q(n), and document rather than silently emend them.

Printed page 257 (physical 11)

A029 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, a black scanner border, and a neighboring-page sliver; it begins with the warning concluding 1.1.18 and ends immediately before Lemma 1.2.4.

Exclude copy matter and sliver text; preserve the complete opening of Section 1.2 and the page break before the lemma.

A030 - NOTATION_CHECK

The authority gives Ψ(x,h′(i)y)=gΨ(g$^{\{-1\}x,h(i)g}${-1}y), uses (2πi)^n, cites results [1] and [8], states the representation as ad μ, and defines G*(R) by g=int(h(i))σ(g).

Preserve the external g-action, exponent n, citations, ad μ, and the exact defining equality in both layers.

D031-N003 - SOURCE OMISSION PRESERVED

In 1.2.3 the authority literally prints $G^*(\mathbf{R}) = \{g \in (\mathbf{C}) \mid g = \mathrm{int}(h(i))\sigma(g)\}$. The returned editions silently inserted $G_{\mathbf{C}}$ before $(\mathbf{C})$. Both staged editions now reproduce the source omission. The apparent mathematical incompleteness is a source fact, not a newly supplied correction.

Printed page 258 (physical 12)

A032 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, a black scanner border, and a neighboring-page sliver; the text starts with Lemma 1.2.4 and ends mid-sentence in 1.2.6.

Exclude copy matter and sliver text; preserve the lemma, every displayed bijection, the root-data formulas, and the unfinished final sentence.

A033 - TRANSLATION_DECISION

The French authority literally prints “groupe de Weil” for W, while the mathematical term in this context is the Weyl group. It also prints Hom(G_m,G)/G_C(C) in the induced bijection, omitting the subscript C on the target Hom.

Keep “groupe de Weil” and the displayed Hom target literal in the frozen French. Render the term as “Weyl group” in English as an explicit terminological translation, but preserve the source’s Hom target without normalization.

Printed page 259 (physical 13)

A035 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, a black scanner border, and a neighboring-page sliver; the article text starts by completing 1.2.6 and ends after the opening of 1.3.2.

Exclude the running head, folio, border, and sliver; preserve the exact continuation from printed 258 and the unfinished final clause.

A036 - NOTATION_CHECK

The authority gives $\pi_0(K) \simeq \pi_0 G(R)$, Centr(h)=$K^+$=K∩G(R)$^+$, identifies the complexification of K^+ with the centralizer of μ_h, and in Corollary 1.2.8(ii) uses all g∈G(R)−G(R)$^+$.

Preserve these four readings in both layers; do not replace them by inclusions, a centralizer of h, or a smaller set of elements.

Printed page 260 (physical 14)

A038 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, a black scanner border, and a neighboring-page sliver; it starts with condition (iii’) and ends mid-sentence after “pour τ”.

Exclude copy matter and preserve the page break, including the unfinished definition of τ.

A039 - SOURCE_ANOMALY

The French authority prints the grammatically irregular phrases “qui est essentiellement équivalent à figurent” and “Ce problème celui résolu”, and cites 1.18(a).

Preserve these source forms in the frozen French; regularize the prose in English without changing the mathematical claim or reference.

D031-N011 - QUALIFIED EQUIVALENCE RETAINED IN ENGLISH

The displaced French phrase “essentiellement equivalent” in the grammatically damaged paragraph after 1.3.2 remains literal in the diplomatic French. The English now reads “This problem is essentially equivalent to the one solved by Satake in [11].” This conservative editorial placement restores the qualification omitted by the returned fluent sentence. It is a documented translation judgment, not a change to a formula or reference.

Printed page 261 (physical 15)

A041 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, a black scanner border, and a neighboring-page sliver; it contains the completion of 1.3.6, Lemma 1.3.7, 1.3.8, and all of Table 1.3.9.

Exclude copy matter and preserve the complete formula and table as article content.

A042 - DIAGRAM_AUDIT

Table 1.3.9 contains the A, B, C, DˆR, DˆH, E_6, and E_7 Dynkin diagrams, with circled special vertices, underlined qualifying vertices, multiple-bond arrows, and every rational label.

Recreate the table natively from authority pixels; audit every node, bond direction, circle, underline, label, and the condition k+2≥5.

A043 - NOTATION_CHECK

The authority states $\langle\mu,-\tau(\alpha)\rangle=\langle\mu,\alpha\rangle-1$ and equation (1.3.6.1) as $\langle\mu,\alpha+\tau(\alpha)\rangle=1$; the table assigns each vertex the number $\langle\mu,\omega\rangle$.

Preserve the pairings and equation number exactly in French and English, and use the same values in the diagram labels.

D031-N004 - SOURCE ROOT-VARIABLE DISCREPANCY PRESERVED

At the opening of printed page 261 the authority uses the pairing $\langle\mu, z\rangle$ but then says r is a root. It also prints the singular French wording "combinaison Z-lineaire de racine". The returned editions silently replaced both variables by γ, and French pluralized "racine". Both staged readers restore the source variables z and r, and the diplomatic French restores the singular wording. The English grammatical phrase "a Z-linear combination of roots" is retained as translation, without concealing the variable discrepancy.

D031-N012 - DYNKIN TABLE LABEL CLEARANCE

A fresh rendered-page audit found that the central labels 2 and 3 in the E6 and E7 diagrams intersected their vertical branch, and that the lower-right label of the quaternionic D diagram touched its circled vertex. Native TikZ label anchors are adjusted: central branch labels move above-right, other above-labels move up 4pt, and right-labels move right 4pt. All vertex locations, bonds, circles, underlines, mathematical values, and label-to-vertex associations remain unchanged. This is a presentation repair, not source normalization.

Printed page 262 (physical 16)

A045 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, a black scanner border, and a neighboring-page sliver; it contains Remarks 1.3.10, the openings of Sections 2 and 2.0, and 2.0.1-2.0.2.

Exclude copy matter and preserve the section transition, exact sequences, quotient action, and final commutator map.

A046 - FAULT_GATE

Condition 2.0.1(a) in the authority is r($\varphi(\gamma)$)=int_γ, expressed in prose as r($\varphi(\gamma)$) being the inner automorphism int_γ of G; there is no extra g.

Enforce the authority reading in the frozen French and English and record candidate fault F02 as repaired.

A047 - DIAGRAM_AUDIT

The authority gives the two exact rows in (2.0.1.1), the isomorphism Γ\G≃Δ(G*_{Γ}Δ), the open subgroup G/Ker(φ), and G=$\widetilde{G}$*_{Z($\widetilde{G}$)}Z(G).

Preserve every object, arrow, quotient, kernel, and amalgamated-product subscript; render and inspect the diagram.

Printed page 263 (physical 17)

A049 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, a black scanner border, and a neighboring-page sliver; it starts with the continuation of 2.0.2 and ends with the hyphenated opening of Remark 2.0.6.

Exclude copy matter and preserve the page-spanning word and all of 2.0.3-2.0.5.

A050 - NOTATION_CHECK

The authority defines Γ_S=$\rho\widetilde{G}$(A_S)∩G(k) inside G(A_S), uses H_*^1 for classes trivial on every cyclic subgroup, uses v∉S in Proposition 2.0.4(ii), and invokes the Hasse principle for $\widetilde{G}$.

Preserve the ambient intersection, starred cohomology, complement-of-S condition, and covering group in both layers.

Printed page 264 (physical 18)

A052 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, a black scanner border, and a neighboring-page sliver; it completes Remark 2.0.6 and runs through Corollary 2.0.13.

Exclude copy matter and preserve the continuation, every displayed containment, and the final corollary.

A053 - FORMULA_AUDIT

In Corollary 2.0.8 the authority uses a compact open K⊂G(A_T) and the displayed expression $\rho(\widetilde{G}$(k) · $\widetilde{G}$(k_{T−S})×ρ^{-1}K); Corollary 2.0.12 uses G^{der}, not a further tilde-derived group.

Preserve the host group of K, every factor and inverse image in the containment, and G^{der} throughout 2.0.11-2.0.13.

Printed page 265 (physical 19)

A055 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, a black scanner border, and a neighboring-page sliver; it begins with Corollary 2.0.14 and runs through 2.1.2.

Exclude the running head, folio, border, and sliver; preserve all owned-page article text and the page boundary after the analytic-space statement.

A056 - QUOTIENT_AND_AXIOM_AUDIT

The authority defines π(G)=G(A)/G(k)$\rho\widetilde{G}$(A), gives axioms (2.1.1.1)-(2.1.1.5), uses strong approximation for $\widetilde{G}$, and writes the Shimura quotient with G(A^f)/K and Γ'*g=gKg^{-1}*$\cap$*G(Q)+*.

Preserve the unclosed denominator in (2.0.15.1), every axiom, the covering-group density statement, and each quotient and subgroup exactly in French and English.

Printed page 266 (physical 20)

A058 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, black scanner border, and neighboring-page sliver; it continues 2.1.2 and ends with Proposition 2.1.7.

Exclude copy matter and sliver text; preserve the continuation and all displays through the proposition.

A059 - FORMULA_AND_ACTION_AUDIT

The authority gives (2.1.3.1)-(2.1.3.2), the right action g:{}*K M_C→{}*{g^{-1}Kg}M_C, the definitions of M_C^0 and τ(G_1), the closure notation superscript minus, and both amalgamated-product identities (2.1.6.1)-(2.1.6.2).

Preserve every closure, quotient, product, action index, completion sign, fiber-product subscript, and equation number; render-inspect all displays.

D031-N005 - SUPERSCRIPT PLUS RESTORED

In the prose of 2.1.6 immediately before (2.1.6.1), the authority prints $\Gamma \subset G_1(\mathbf{Q})^+$ with a superscript plus. Both returned editions placed the plus in a subscript. The staged readers restore the superscript only in that prose occurrence; the distinct subscript plus in (2.1.6.2) is preserved.

D031-N007 - SOURCE ADELE ARGUMENT RESTORED

In the same prose paragraph of 2.1.6, the source defines $\Gamma = \rho\widetilde{G}_0(A) \cap G_1(\mathbf{Q})$ using A, with no superscript f. Both returned editions silently used A^f. The staged readers restore the literal source argument A in that definition only; nearby uses of A^f are unchanged.

Printed page 267 (physical 21)

A061 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, black scanner border, and neighboring-page sliver; it begins by completing Proposition 2.1.7 and ends mid-sentence in 2.1.13.

Exclude copy matter and sliver text; preserve both page-boundary continuations.

A062 - SOURCE_ANOMALY_AND_ACTION_AUDIT

At the page opening the authority literally prints Γ\X^0, then defines connected Shimura varieties, the equality M_C$^{0(G,X)=\text{M_C}}$0(G^{ad,G}der,X^+), the closure Z(Q)^-, the quotient description of M_C, and the int(γ) action of G^ad(Q)_1.

Preserve the printed X^0 in the frozen French and translate it literally; retain every closure, quotient, group subscript, and action arrow from authority pixels.

Printed page 268 (physical 22)

A064 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, black scanner border, and neighboring-page sliver; it completes 2.1.13, contains all of 2.1.14-2.1.16 and 2.2.1-2.2.2, and ends in 2.2.3.

Exclude copy matter and sliver text; preserve the opening and closing continuations and all intervening displays.

A065 - ACTION_NORM_AND_REFLEX_FIELD_AUDIT

The authority gives the two fiber products in (2.1.13.1), the single equality annotated 2.0.13 in (2.1.15.1), the Summary 2.1.16 action, μ_h:G_m→G_C, the reflex-field inclusion, and NR_{E′}(μ)=NR_E(μ)∘N_{E′/E}.

Preserve the exact denominators and amalgamation bases, use one annotated equality only, and retain the μ, reflex-field, norm, and composition notation in both layers.

Printed page 269 (physical 23)

A067 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, black scanner border, and neighboring-page sliver; it begins with the continuation of 2.2.3 and ends after the opening sentence of 2.2.6.

Exclude copy matter and sliver text; preserve the exact page-spanning continuations.

A068 - CANONICAL_MODEL_CONDITION_AUDIT

The authority defines r_E(T,X) as the inverse reciprocity composite, acts by right translation, defines the special-point action r, and states canonical-model conditions (a),(b) and weak condition (b*) with E(τ) and the displayed scalar-extension isomorphism.

Preserve the inverse convention, right action, representative class, Galois subgroup inclusion, E(τ), intersection only in (b*), and the complete equivariant isomorphism.

Printed page 270 (physical 24)

A070 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, black scanner border, and neighboring-page sliver; it completes 2.2.6, opens 2.3, and ends immediately after “La définition 2.2.1 donne:”.

Exclude copy matter and sliver text; preserve both page-boundary continuations and all owned 2.3 text.

A071 - CRITERION_AND_STRUCTURE_AUDIT

The authority gives Criterion 2.3.1, Proposition 2.3.2, Corollary 2.3.3, the symbols Ψ and $\mathfrak{S}$^±, G_R/w(G_m), the subgroup G_2, G=R_{F/Q}G^s, the Dynkin data D and K_D, and Σ(X).

Preserve every embedding arrow, type, group, field, product decomposition, Dynkin symbol, and terminal Σ(X); check the English mathematics independently against the scan.

Printed page 271 (physical 25)

A073 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, a black scanner border, and a neighboring-page sliver; it begins with Proposition 2.3.6 and ends within the D_l^H subcase of Table 2.3.8.

Exclude the running head, folio, border, and sliver; preserve the exact opening and the unfinished final clause at the page boundary.

A074 - CLASSIFICATION_AND_COVER_AUDIT

The authority gives (2.3.7.1), the kernel diagram with G_nc and Ker(G_1R to G_c)^0, conditions (a)-(d), the notation Sigma(X) and G-tilde(S), and the A, B, C, D_l^R, D_l^H cases of Table 2.3.8.

Preserve every group, kernel, Galois-stability condition, underlined-vertex criterion, cover, case label, and page-spanning continuation in both language layers.

Printed page 272 (physical 26)

A076 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, scanner border, and neighboring-page sliver; it completes Table 2.3.8, gives 2.3.9 and most of Proposition 2.3.10, and ends mid-sentence.

Exclude copy matter and sliver text; preserve the opening continuation, every D_4 subcase, and the unfinished closing line.

A077 - D4_AND_HODGE_CONSTRUCTION_AUDIT

The authority literally uses sections tau of X to I in the D_4 discussion, distinguishes D_4^R and D_4^H, excludes D_4^R from the redefined D_4^H case, defines h_T, and states Proposition 2.3.10 with its reflex field and derived-cover conditions.

Retain the literal X to I notation, all Sigma(X) containments, the D_4^R and D_4^H logic, the Hodge types, and the exact covering and field statements.

Printed page 273 (physical 27)

A079 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, scanner border, and neighboring-page sliver; it continues the definition of V(s), completes Proposition 2.3.10, and ends after Remark 2.3.11(c).

Exclude copy matter and preserve both page-boundary continuations and all intervening representation and Hodge-structure text.

A080 - BRAUER_MODULE_AND_HODGE_AUDIT

The authority uses the unindexed direct sums of V(s) and V_s, the Brauer obstruction, the quotient G-tilde-prime, the K_S-module structure, h_2 and h_3, the tensor decomposition of K tensor_F V, and G_3 generated by K^* and G_2.

Preserve the direct-sum signs, faithful quotient, fractional types, generator groups, tensor factors, reflex fields, and the transition to Remark 2.3.11.

Printed page 274 (physical 28)

A082 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, scanner border, and neighboring-page sliver; it begins with Remark 2.3.11(d), opens Section 2.4, and ends in the final clause of the Galois-descent paragraph.

Exclude copy matter and preserve the opening and closing continuations, the section heading, and all displayed morphisms.

A083 - REFLEX_FIELD_AND_DESCENT_AUDIT

The authority gives E(K_S-prime-star,h), Remarks 2.3.12-2.3.13, the norm and q_M morphisms (2.4.0.1)-(2.4.0.2), the SGA 4 one-half citation, and the three explicit chi descent conditions.

Preserve the prime-and-star placement, all field and type statements, both numbered arrows, and the literal descent isomorphisms and compatibility.

Printed page 275 (physical 29)

A085 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, scanner border, and neighboring-page sliver; it completes 2.4.1 and contains Example 2.4.2 through all of 2.4.5.

Exclude copy matter and preserve the opening continuation, commutator formulas, torsor conventions, exact sequence, equivalences, and Picard-category statement.

A086 - COMMUTATOR_TORSOR_AND_EXACT_SEQUENCE_AUDIT

The authority maps the commutator to G-tilde(k), uses the trivial torsors G_d, G_1d, and G_2d, defines H^0 and H^1 of [G_1 to G_2], and gives the exact sequence (2.4.3.1) and the kernel-category equivalences.

Preserve the tilde codomain, commutator order, right-action convention, every torsor subscript, category bracket, kernel, arrow, and exact-sequence term.

Printed page 276 (physical 30)

A088 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio; it is a tightly cropped single-page image without a neighboring-page sliver or scanner border. It contains 2.4.6 and most of 2.4.7 and ends mid-sentence after equation (2.4.7.3).

Exclude copy matter and preserve the unfinished final sentence and all trace, norm, equivalence, and Picard-category material.

A089 - TRACE_NORM_AND_PICARD_DIAGRAM_AUDIT

The authority literally prints H^1(k,R_{k prime/k}G) without a prime on the final G, gives the additive trace functor (2.4.6.1), the plain-arrow equivalence (2.4.7.1), decomposition x_i=rho(g_i)z_i, and equations and diagram (2.4.7.2)-(2.4.7.3).

Preserve the literal Shapiro target, all trace and norm indices, the plain displayed functor arrow, decomposition order, vertical and diagonal diagram arrows, and final equality.

D031-N006 - COPY-MATTER DESCRIPTION CORRECTED

The inherited A088 source-fact description incorrectly claimed a scanner border and neighboring-page sliver on physical page 30. Independent inspection shows a tightly cropped single-page image with running head and folio but no such sliver or border. A088 is corrected in this normalized apparatus. Article text and mathematics are unchanged.

Printed page 277 (physical 31)

A091 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, scanner border, and neighboring-page sliver; it completes 2.4.7, contains Proposition 2.4.8 and 2.4.9, and ends after the adelic norm morphism.

Exclude copy matter and sliver text; preserve the opening continuation, every displayed diagram and morphism, and the complete page boundary.

A092 - NORM_COHOMOLOGY_AND_SOURCE_ANOMALY_AUDIT

The authority gives the action formula g_1+g_2=g_1 int_{x_1}(g_2), the norm functor (2.4.7.4), Proposition 2.4.8 with G-tilde-prime in its prose but unprimed G-tilde in the upper-left diagram term, plain Z(C) and Z(R) in the archimedean diagram, the literal denominator G(V prime)/rho G-tilde(V), and the adelic norm morphism.

Preserve every prime, tilde, center, norm index, action, arrow, and the source-local unprimed V in the denominator without normalization.

D031-N008 - VERTICAL INJECTION HOOKS RESTORED

Both vertical arrows in source diagram (2.4.8.1) have hooked injection tails. The returned native diagrams used ordinary vertical arrows. Both staged readers restore the two hooks, preserving their downward directions and all source objects and labels. The separate following norm diagram has ordinary vertical arrows and is unchanged.

Printed page 278 (physical 32)

A094 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, scanner border, and neighboring-page sliver; it opens Construction 2.4.10, gives both methods and their first diagrams, and ends mid-sentence.

Exclude copy matter and preserve the opening transition, both construction methods, all diagrams, and the unfinished closing line.

A095 - DESCENT_DIAGRAM_AND_CENTRALIZER_AUDIT

The authority gives diagrams (1)_m and (1), the literal centralizer factor (rho inverse of T)^0, the equivalence labeled 2.4.4, and the formula m prime(t)=(rho((g,m(t))))m(t) with its outer parentheses.

Preserve every group, centralizer, vertical and horizontal arrow, equivalence label, parenthesis, and descent datum exactly as witnessed.

Printed page 279 (physical 33)

A097 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio; it is a tightly cropped single-page image without a neighboring-page sliver or scanner border. It continues Construction 2.4.10, contains its additivity diagram and 2.4.11, and ends mid-sentence.

Exclude copy matter and preserve the opening continuation, every diagram and label, and the unfinished final phrase.

A098 - SOURCE_ANOMALY_AND_ADDITIVITY_DIAGRAM_AUDIT

The authority literally begins with rho((m,m(t))) rather than rho((g,m(t))), then gives the complete (2.4.10.1) and additivity diagrams, the product G-tilde times Z^0, and the scalar-extension and norm-compatibility diagrams.

Retain the source-local m anomaly, all oblique and vertical arrows, Z^0, every prime, and the complete diagram structures.

D031-N009 - COPY-MATTER DESCRIPTION CORRECTED

The inherited A097 source-fact description incorrectly claimed a scanner border and neighboring-page sliver on physical page 33. Independent inspection shows a tightly cropped single-page image with running head and folio but no such sliver or border. A097 is corrected in this normalized apparatus. Article text and mathematics are unchanged.

Printed page 280 (physical 34)

A100 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, scanner border, and neighboring-page sliver; it completes 2.4.11, contains 2.4.12, opens Section 2.5, and ends mid-sentence in 2.5.1.

Exclude copy matter and preserve both page-boundary continuations, all numbered formulas, and the exact opening of Section 2.5.

A101 - DESCENT_NORM_AND_CANONICAL_EXTENSION_AUDIT

The authority literally uses k^I and [G-tilde to G]^I in the descent sentence, unprimed H^1(G-tilde) in 2.4.12, equations (2.4.12.1)-(2.4.12.2), and the quotients, closures, and amalgamated product beginning 2.5.1.

Preserve the exponent I, cohomology group, plus-components, closures, quotient order, and every amalgamated-product datum.

Printed page 281 (physical 35)

A103 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, scanner border, and neighboring-page sliver; it completes 2.5.1, contains 2.5.2 and most of 2.5.3, and ends immediately before the next extension diagram.

Exclude copy matter and preserve the opening continuation, all exact sequences and diagrams, and the terminal page boundary.

A104 - EXTENSION_FUNCTORIALITY_AND_MISSING_SYMBOL_AUDIT

The authority gives extension (2.5.1.4), the two 2.5.2 diagrams, N_{E/Q} q_M, and the literal phrase "Nous noterons (G,M) son inverse" with no printed r, while the following clause uses r(G,M) and defines the inverse-image extension.

Retain the exact extension and diagrams, the missing-symbol source fact, the subsequent r(G,M), and all E-prime_E notation without silent repair.

Printed page 282 (physical 36)

A106 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, scanner border, and neighboring-page sliver; it begins with the continuation extension diagram, contains 2.5.4 through Definition 2.5.7, and ends after the field-change diagram.

Exclude copy matter and preserve the opening continuation, every construction and definition, and both rows of the closing diagram.

A107 - CENTRAL_EXTENSION_COPRODUCT_AND_FIELD_CHANGE_AUDIT

The authority states M_1 isomorphic to M, Lemma 2.5.5, the coproduct formula for Z, Construction 2.5.6, Definition 2.5.7, and a field-change sentence naming the corresponding extension with M while its display uses M_F.

Preserve every cover, lift, kernel, coproduct, fiber product, completion, norm arrow, and the literal prose-versus-display M and M_F distinction.

Printed page 283 (physical 37)

A109 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, scanner border, and neighboring-page sliver; it completes the field-change discussion, contains 2.5.8-2.5.10, and ends after diagram (2.5.10.1).

Exclude copy matter and preserve the opening continuation, both page-local extension diagrams, equations (2.5.8.1) and (2.5.9.1), and the complete terminal diagram.

A110 - EXTENSION_QUOTIENT_AND_UNIVERSAL_CASE_AUDIT

The authority distinguishes G-prime from G-tilde throughout 2.5.8, uses G-ad(Q) in (2.5.9.1), and in the universal-case sentence retains the prose extension E_E(G,G-prime,M), the ambient E_{E(G,X)}(G,G-prime,X), and the quotient E_E(G,G-prime,X) of E_E(G,G-tilde,X).

Preserve every cover, relative completion, vertical norm arrow, M-versus-X occurrence, field product, and page-boundary continuation without normalization.

D031-N010 - ADDED DIAGRAM ARROW REMOVED

The first, unnumbered diagram of 2.5.8 has no horizontal arrow from the lower middle ellipsis to the lower right $\pi_0\pi(T)$ in the authority. Both returned diagrams inserted that arrow. The staged readers remove only this added horizontal arrow and preserve the upward arrow. The lower-row horizontal arrow in the subsequent numbered diagram (2.5.8.1) is printed in the source and is retained.

Printed page 284 (physical 38)

A112 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, scanner border, and neighboring-page sliver; it contains Section 2.6 through most of 2.6.4 and ends immediately before the first functoriality diagram.

Exclude copy matter and preserve the section heading, sign convention, equations (2.6.1.1)-(2.6.2.1), Theorem 2.6.3, and the unfinished closing line.

A113 - RECIPROCITY_SIGN_AND_INVERSE_THEOREM_AUDIT

The authority states that the left action of sigma coincides with the right action of r_{G,X}(sigma), with no inverse. Separately, Theorem 2.6.3 states that the reciprocity morphism itself is the inverse of pi_0 N_{E/Q} q_M.

Enforce the no-inverse action convention while preserving the distinct inverse statement in the theorem, all bars and pi_0 symbols, the torus inclusion iota, and every norm field.

Printed page 285 (physical 39)

A115 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, scanner border, and neighboring-page sliver; it completes 2.6.4, opens Section 2.7, contains 2.7.1-2.7.3, and ends immediately before the induction quotient formula.

Exclude copy matter and preserve both opening norm diagrams, the section transition, every action convention, and the page-spanning induction sentence.

A116 - NORM_FUNCTORIALITY_AND_INDUCTION_SETUP_AUDIT

The authority gives two distinct norm diagrams with q_mu and q_M, the composite N_{E(G,X)/Q} q_M N_{E(T,h)/E(G,X)}, the quotient conventions K-backslash S and S/K, stabilizer Delta, and delta(gamma,t)=(gamma delta^{-1},delta t).

Preserve all fields, diagram arrows, equality marks, left and right quotient order, the Delta action, and the unfinished page boundary.

Printed page 286 (physical 40)

A118 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, scanner border, and neighboring-page sliver; it begins with the induction quotient formula, contains 2.7.4-2.7.9, and opens 2.7.10(a).

Exclude copy matter and preserve the opening formula, Galois-descent lemma, extension diagram, action identity, and the full first clause of the connected-model definition.

A119 - INDUCTION_EXTENSION_AND_GALOIS_ACTION_AUDIT

The authority has the coproduct indexed by gamma in K-backslash Gamma/Delta=K-backslash pi with term (gamma K gamma^{-1} intersect Delta)-backslash T; the lower row of the extension diagram begins with Gamma, and 2.7.7 states sigma dot x=x dot r(sigma) and that script E does not depend on e.

Preserve the conjugation order, double-coset index, lower-row Gamma, script-E statement, and the action formula with no inverse.

Printed page 287 (physical 41)

A121 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, scanner border, and neighboring-page sliver; it completes 2.7.10, contains 2.7.11-2.7.13, and ends at the hyphenated first half of “canonique.”

Exclude copy matter and preserve the opening continuation, all functoriality conditions, the amalgamated action, Proposition 2.7.13, and the exact terminal word break.

A122 - CONNECTED_MODEL_AND_AMALGAMATED_ACTION_AUDIT

The authority uses the general cover G-prime in 2.7.10-2.7.11, gives the two amalgamated products with G-ad(Q)*+ versus G-ad(Q)ˆ+, defines pi using pi_0(Mˆ0*{Qbar}(G,X))=pi_0(M_C(G,X)), and then uses the natural projection of M_{Qbar}(G,X).

Preserve the prime cover, both plus placements and amalgamation subscripts, the source-local superscript-zero distinction, every Galois intersection, and the cross-page "cano- / nique" continuation.

D031-N002 - COMPLEX-BASE SUBSCRIPTS RESTORED

Independent source-pixel inspection of 2.7.11(a)-(b) identified three missing complex-base subscripts in both returned editions: the individual factors and their product in (a), and the quotient target in (b). Both staged readers restore $M^0_{\mathbf{C}}$ in precisely these three expressions. The initial target in (b), which has no complex-base subscript in the source, is unchanged.

Printed page 288 (physical 42)

A124 - COPY_MATTER_EXCLUSION

The authority leaf has a repeated running head and folio, scanner border, and neighboring-page sliver; it begins with the second half of "canonique," contains 2.7.14-2.7.19, and ends after the opening line of Theorem 2.7.20.

Exclude copy matter and preserve the cross-page opening, all lifting and existence results, and the exact terminal boundary without importing page 289.

A125 - LIFTING_DICTIONARY_AND_EXISTENCE_CRITERIA_AUDIT

The authority writes the lifted morphisms in 2.7.14 and Lemma 2.7.16 as maps from S into G, gives h(z)=mu(z) bar-mu(bar-z), uses a G_1(R)ˆ0-conjugacy class, states Corollaries 2.7.18-2.7.19, and begins Theorem 2.7.20 with G Q-simple adjoint and G-prime a covering.

Preserve the literal target G, both complex-conjugation bars, the superscript 0, all field and linear-disjointness clauses, and stop exactly at the owned page boundary.

Printed page 289 (physical 43)

A127 - COPY_MATTER_EXCLUSION

The terminal authority leaf has the repeated running head and folio, a large neighboring-page sliver, scanner borders, a blank remainder after the address, and physical EOF.

Exclude the running head, folio, sliver, borders, and blank remainder from both reader bodies; retain the terminal address; record that the exact witness ends here and that no page 290 exists.

A128 - THEOREM_CONTINUATION_AND_COROLLARY_AUDIT

The authority completes Theorem 2.7.20 with cases A, B, C, DˆR, and DˆH using G-prime, then gives Corollary 2.7.21 with M(G,X), the system (G_i,X_iˆ+), and the covering Gˆder of Gˆad as a quotient of the product of the indicated coverings.

Preserve the source-local G-prime notation, DˆR and DˆH without an added subscript, the singular French 'morphisme' and 'système', M(G,X) without a C-subscript, and the exact $G^{der/G}$ad quotient relation.

A129 - BIBLIOGRAPHY_AND_TERMINAL_MATTER_AUDIT

The authority prints Bibliographie items 1-13, item 5 as Exposé 389, the repeated-author rules in items 6, 7, and 13, the SGA line with SGA4ˆ{1/2}, and the terminal IHES address.

Replay all thirteen entries in order; enforce Exposé 389; preserve titles, journal data, repeated-author rules, SGA numbering, and the literal terminal address before the blank remainder and EOF.